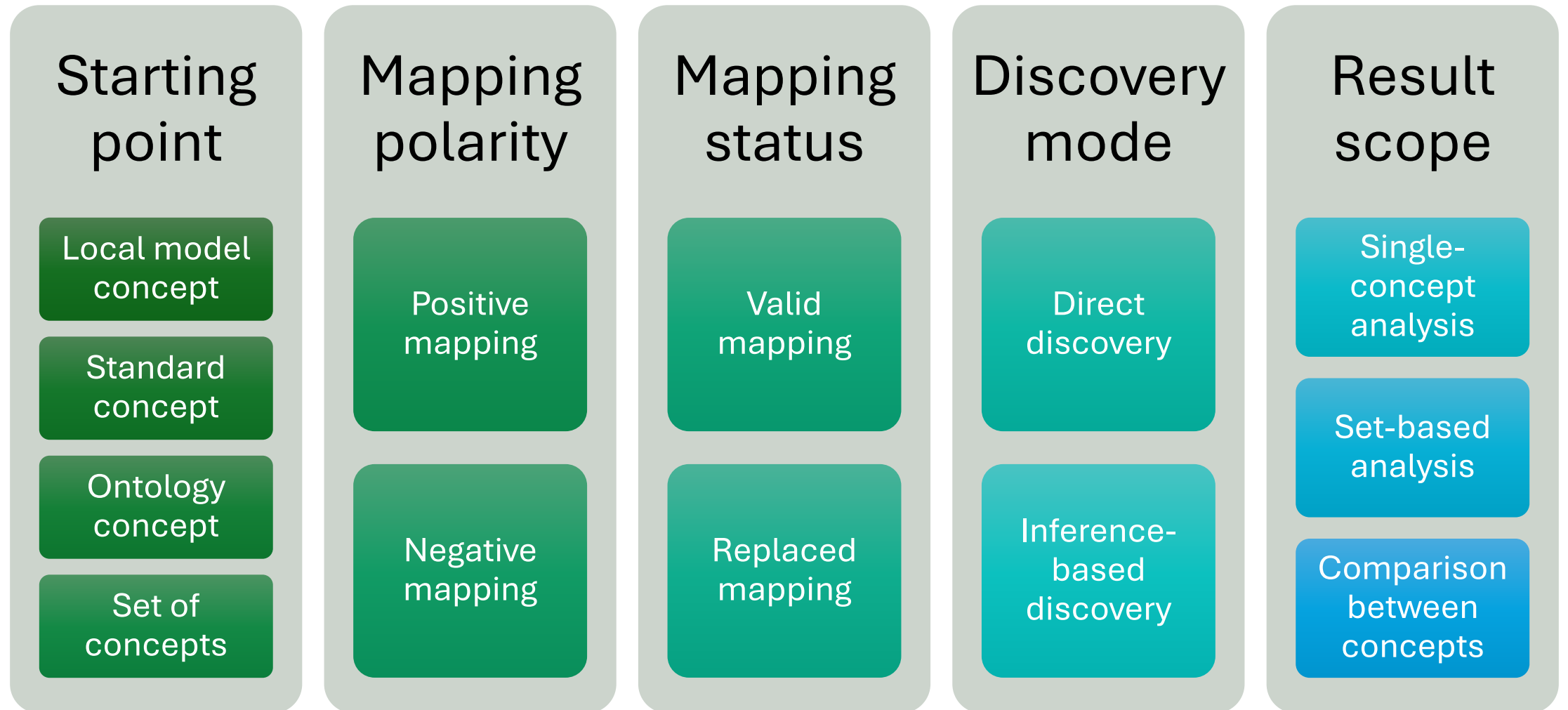


Use Scenarios for the Health-RI Semantic Interoperability Initiative

Considered Axis



Scenario categories

S: Selection

- Scenarios in which the main goal is to identify the correct ontology meaning before creating a mapping.

D: Direct discovery

- Scenarios in which related concepts are retrieved directly from existing mappings to the same ontology concept.

I: Inference-based discovery

- Scenarios in which additional relations become available only after applying the inference rules over the mapping graph.

L: Lifecycle and governance

- Scenarios concerned with replaced mappings, conflict detection, and other cases that require interpretation of mapping history or governance review.

C: Comparison

- Scenarios in which multiple concepts are analyzed together in order to identify overlap, difference, or non-alignable cases.

Users considered

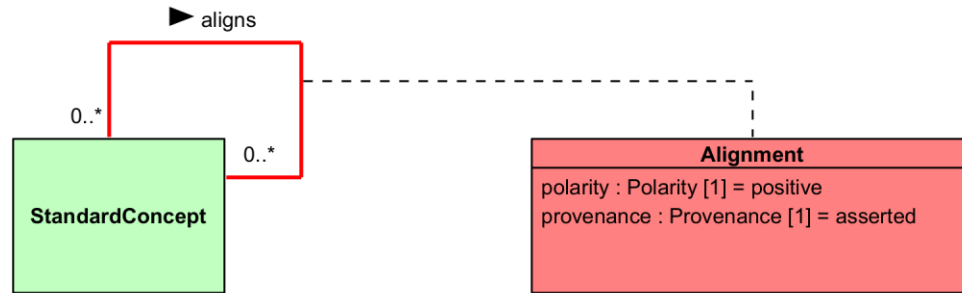
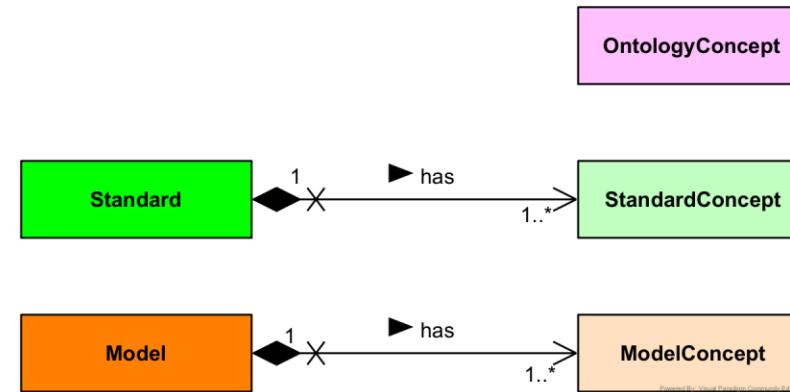
Model Owner

- A user responsible for local model concepts, such as dataset fields, columns, or internal data elements.
- This user typically needs to:
 - choose the correct ontology meaning for a local concept
 - map a local concept to the ontology
 - discover related concepts after mapping
 - compare local concepts across semantic meanings

Standard Owner

- A user responsible for standard concepts and their semantic grounding.
- This user typically needs to:
 - define or refine the ontology grounding of a standard concept
 - discover aligned and non-aligned standard concepts
 - analyze inferred relations created through the rule set
 - review replacement paths and conflicts in standard mappings

Support Model (1/2)



R1. Symmetry of Alignment

Statement:

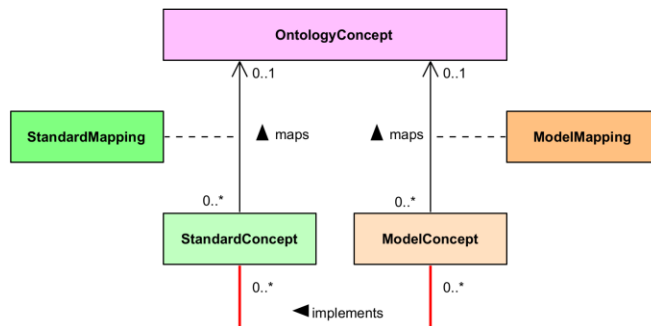
If one StandardConcept aligns with another StandardConcept, then the second StandardConcept also aligns with the first StandardConcept.

Formal rule:

```

aligns(StandardConcept1, StandardConcept2)
<-> aligns(StandardConcept2, StandardConcept1)

```



R2. Propagation of a ModelConcept Mapping to its Implemented StandardConcept

Statement:

If a ModelConcept implements a StandardConcept, and the ModelConcept maps to an OntologyConcept, then the implemented StandardConcept also maps to the same OntologyConcept.

Formal rule:

```

implements(ModelConcept1, StandardConcept1)
and maps(ModelConcept1, OntologyConcept1)
-> maps(StandardConcept1, OntologyConcept1)

```

Inference metadata:

The inferred StandardMapping must have provenance = inferred.

R3. Propagation of a StandardConcept Mapping to its Implementing ModelConcept

Statement:

If a ModelConcept implements a StandardConcept, and the StandardConcept maps to an OntologyConcept, then the implementing ModelConcept also maps to the same OntologyConcept.

Formal rule:

```

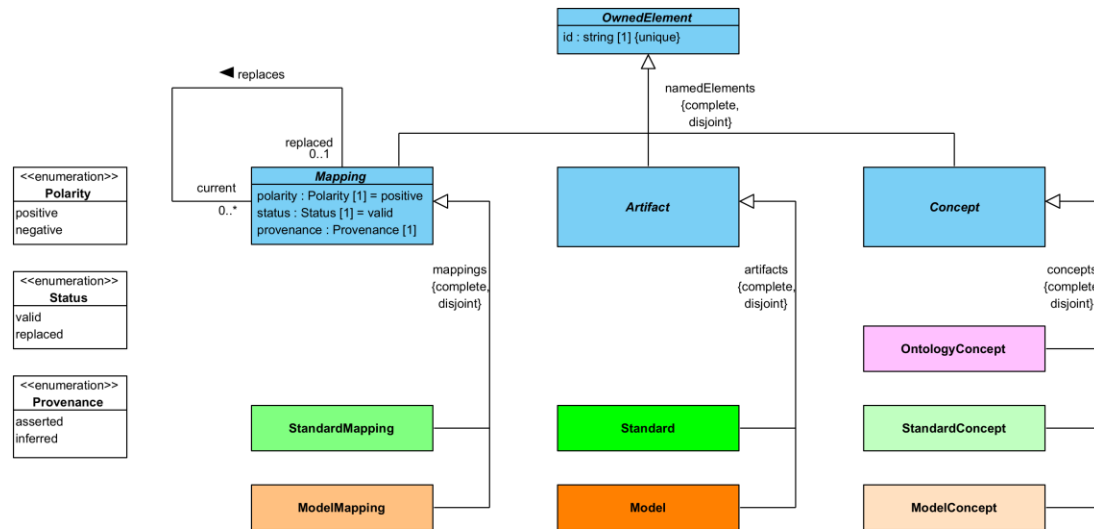
implements(ModelConcept1, StandardConcept1)
and maps(StandardConcept1, OntologyConcept1)
-> maps(ModelConcept1, OntologyConcept1)

```

Inference metadata:

The inferred ModelMapping must have provenance = inferred.

Support Model (2/2)



R4. Alignment Inferred from a Shared Implementing ModelConcept

Statement:
If the same ModelConcept implements two distinct StandardConcept instances, then those two StandardConcept instances are aligned.

Formal rule:
 implements(ModelConcept1, StandardConcept1)
 and implements(ModelConcept1, StandardConcept2)
 and StandardConcept1 != StandardConcept2
 -> aligns(StandardConcept1, StandardConcept2)

Inference metadata:
The inferred Alignment must have provenance = inferred.

R5. Propagation of a StandardConcept Mapping Across an Alignment

Statement:
If one StandardConcept aligns with another StandardConcept, and the first StandardConcept maps to an OntologyConcept, then the second StandardConcept also maps to the same OntologyConcept.

Formal rule:
aligns(StandardConcept1, StandardConcept2)
and maps(StandardConcept1, OntologyConcept1)
-> maps(StandardConcept2, OntologyConcept1)

Inference metadata:
The inferred StandardMapping must have provenance = inferred.

R6. Alignment Inferred from a Shared OntologyConcept Mapping

Statement:
If two distinct `StandardConcept` instances both map to the same `OntologyConcept`, then those two `StandardConcept` instances are aligned.

Formal rule:
 maps(StandardConcept1, OntologyConcept1)
 and maps(StandardConcept2, OntologyConcept1)
 and StandardConcept1 != StandardConcept2
 -> aligns(StandardConcept1, StandardConcept2)

Inference metadata:
The inferred Alignment must have provenance = inferred.

R7. Propagation of a Negative StandardConcept Mapping Across an Alignment

Statement:
If a StandardConcept negatively maps to an OntologyConcept, and that StandardConcept aligns with another StandardConcept, then the other StandardConcept also negatively maps to the same OntologyConcept.

Formal rule:
aligns(StandardConcept1, StandardConcept2)
and maps(StandardConcept1, OntologyConcept1)
and polarity(StandardConcept1, OntologyConcept1) = negative
-> maps(StandardConcept2, OntologyConcept1)
and polarity(StandardConcept2, OntologyConcept1) = negative

Inference metadata:
The inferred StandardMapping must have provenance = inferred.

R8. Incompatibility Between Negative and Positive Mappings to the Same Ontology Concept

Statement:
Two StandardConcept instances must not align if one negatively maps to an OntologyConcept and the other positively maps to the same OntologyConcept.

Forbidden condition:
aligns(StandardConcept1, StandardConcept2)
and maps(StandardConcept1, OntologyConcept1)
and polarity(StandardConcept1, OntologyConcept1) = negative
and maps(StandardConcept2, OntologyConcept1)
and polarity(StandardConcept2, OntologyConcept1) = positive

Use Scenarios Cards

S1. As a **model owner**, I want to check the possible meanings of a new concept before I map it.

This scenario concerns the creation of a new local model concept whose intended meaning is not yet fixed. Because the concept is still unmapped, no related model or standard concepts can be retrieved from the knowledge base on its behalf. The first step is therefore to consult the ontology and identify the candidate ontology concepts that may correspond to the intended meaning of the new local concept. Once the appropriate ontology concept has been selected, a mapping can be created. After that mapping exists, the local concept becomes connected to the ontology-based semantic structure of the knowledge base and can participate in the retrieval of directly and indirectly related concepts.

AS IS

A model owner wants to create a new local concept, but the label is unclear and may have more than one meaning.

TO BE

The owner has found the right ontology concept and can create the mapping.

S1. As a model owner, I want to check the possible meanings of a new concept before I map it.

EXAMPLE

A data steward is designing a new column for a clinical registry and wants to call it “Patient.” Before creating the mapping, they realize that this word can mean different things depending on the context. In one dataset, it may mean a person currently receiving treatment. In another, it may mean a person enrolled in a study. In yet another, it may simply mean the person whose information is recorded in the system.

The steward searches the ontology first. The search returns several candidate ontology concepts, each with a slightly different meaning. After reading the definitions and checking the intended use of the new column, the steward decides that in this dataset “Patient” should mean “person receiving care.” They then create the mapping from the local concept to that ontology concept, knowing that this will later allow them to retrieve related model and standard concepts connected to the same meaning.

S2. As a **standard owner**, I want to check the possible meanings of a new standard concept before I map it.

This scenario concerns the creation of a new standard concept whose intended meaning is not yet fixed. Because the concept is still unmapped, no related concepts can yet be retrieved for it from the knowledge base. The first step is therefore to consult the ontology and review the candidate ontology concepts that may correspond to the intended meaning of the new standard concept. After the appropriate ontology concept has been selected, a mapping can be created. Once that mapping exists, the standard concept becomes part of the ontology-grounded semantic structure of the knowledge base and can be used for the retrieval of directly and indirectly related concepts.

AS IS

A standard owner wants to create a new standard concept, but the label is unclear and may have more than one meaning.

TO BE

The owner has found the right ontology concept and can create the mapping for the new standard concept.

S2. As a **standard owner, I want to check the possible meanings of a new standard concept before I map it.**

EXAMPLE

A standards team is preparing a new concept for a standard they maintain, and the proposed label is “Patient.” Before they publish anything, they want to make sure they choose the right ontology meaning. They know that the same label can be interpreted differently across care, research, and data-management contexts.

They search the ontology and review several candidate concepts. One means “person receiving care,” another means “research participant,” and another refers more generally to a person who is the subject of a health record. After comparing the definitions, examples, and intended use of the new standard concept, the team selects the ontology concept that best matches what they want to standardize. They then create the mapping, making the new standard concept discoverable through that ontology meaning.

D1. As a **model owner**, I want to find all concepts related to my concept after I map it to the ontology.

This scenario concerns the discovery of concepts related to a local model concept that already has a current mapping to an ontology concept. The mapped ontology concept functions as the semantic anchor for retrieval. On that basis, the system can retrieve all model concepts and standard concepts that are also connected to the same ontology concept. The results should be organized by polarity. Positive mappings indicate concepts that express or support the same ontology-grounded meaning, whereas negative mappings indicate concepts that are explicitly incompatible with that meaning. The purpose of the scenario is to enable the identification of semantically related concepts after a local concept has been mapped.

AS IS

A model owner already has a local concept that is mapped to one ontology concept and wants to find other concepts related to it through that same ontology meaning.

TO BE

The owner can see all other concepts connected to the same ontology concept, separated by positive and negative meaning.

D1. As a **model owner**, I want to find all concepts related to my concept after I map it to the ontology.

EXAMPLE

A team has already created a local concept called “Patient” in one dataset and mapped it to the ontology concept “person receiving care.” Now they want to understand how this concept relates to the rest of the knowledge base. Instead of looking only at the local model, they ask the system to show everything else that is connected to that same ontology meaning.

The system returns several results. It shows standard concepts from different standards that are also mapped positively to “person receiving care,” which suggests they express the same or very similar meaning. It also shows a few concepts mapped negatively to that ontology concept, which tells the team that those concepts should not be treated as equivalent. Finally, it shows other local model concepts that are linked to the same ontology concept, helping the team see whether a similar concept already exists elsewhere in the organization.

D2. As a model owner, I want to see what follows when my concept is explicitly marked as not matching one ontology meaning.

This scenario concerns relation discovery starting from a local model concept that has a current negative mapping to an ontology concept. Although the mapping is negative, the ontology concept still functions as a valid semantic anchor for retrieval. On that basis, the system can retrieve model concepts and standard concepts that are also connected to the same ontology concept and organize the results by polarity. Concepts with positive mappings represent a meaning that the seed concept explicitly does not take and therefore must not be treated as semantically equivalent to it for that interpretation. Concepts with negative mappings indicate other concepts that likewise reject that ontology meaning. These concepts are relevant for further analysis, but they must not be treated as automatically equivalent to the seed concept. The scenario therefore formalizes how a negative local mapping can be used to identify excluded meanings and to support the comparative analysis of concepts that reject the same ontology-grounded interpretation.

AS IS

A model owner has a local concept with a current negative mapping to an ontology concept and wants to see what other concepts are connected to that same ontology meaning.

TO BE

The owner can see all other concepts connected to the same ontology concept, separated into concepts that are also negative and concepts that are positive.

D2. As a **model owner**, I want to see what follows when my concept is explicitly marked as not matching one ontology meaning.

EXAMPLE

A data steward is modeling a local column called “Patient,” but after reviewing the dataset documentation, it becomes clear that the column is not being used to mean a person receiving treatment. In this dataset, the column refers to a different kind of subject, so the steward records a negative mapping from the local concept to the ontology concept “person receiving care.”

The steward then queries the system to understand the semantic implications of that decision. The system returns standard concepts and local concepts that are positively mapped to “person receiving care,” making it clear which concepts do use that meaning and therefore must not be treated as equivalent to the local concept being modeled.

The system also returns concepts that are negatively mapped to the same ontology concept. These results do not imply that those concepts are automatically equivalent to the seed concept. Instead, they show that those concepts reject the same ontology meaning and may therefore be worth further comparison. This helps clarify both what the local concept does not mean and which other concepts exclude the same interpretation.

I1. As a **standard owner**, I want to start from a positive mapping and see both what aligns with my concept and what should not align with it.

This scenario concerns relation discovery starting from a standard concept that has a current positive mapping to an ontology concept. The positively mapped ontology concept serves as the initial semantic anchor for retrieval. Standard concepts that are also positively mapped to that ontology concept align with the seed concept under the rule set, whereas standard concepts that are negatively mapped to the same ontology concept must not be treated as semantically equivalent and must not align with it. In addition to direct retrieval, this scenario also supports indirect expansion. If another positively related standard concept also has a positive mapping to a different ontology concept, that additional ontology concept becomes relevant through ontology-based closure. The results should therefore be grouped by ontology concept, since the same concept may appear in more than one branch of the discovered semantic neighborhood.

AS IS

A standard owner has a standard concept with a current positive mapping to an ontology concept and wants to see which other standard concepts also map positively to that ontology meaning and which ones map negatively to it.

TO BE

The owner can retrieve positively related concepts that align with the seed concept, negatively related concepts that should not align with it, and additional concepts that become reachable indirectly through ontology-based closure.

I1. As a **standard owner**, I want to start from a positive mapping and see both what aligns with my concept and what should not align with it.

EXAMPLE

A standards team is reviewing their concept “Patient,” which is positively mapped to the ontology concept “person receiving care.” Their first question is straightforward: which other standard concepts also use that same ontology meaning? Those concepts are important because they align with the team’s concept for that meaning and may be candidates for reuse, harmonization, or comparison.

The system first shows the concepts that also map positively to “person receiving care.” It also shows the concepts that map negatively to that same ontology concept, because those should not be treated as the same meaning and should not align with the seed concept.

The team then notices that one of the positively related standard concepts is also positively mapped to another ontology concept, such as “person in a service process.” Because of the inference rules, that second ontology meaning also becomes relevant for the original concept. The system therefore opens a second result branch for that ontology concept and shows additional standard concepts that map positively to it, as well as concepts that map negatively to it. This helps the team understand both what aligns with their concept and what should be kept separate, while keeping the results organized by ontology meaning rather than mixing everything in one flat list.

D3. As a standard owner, I want to start from a negative mapping and see both what my concept cannot align with and what other concepts reject that same ontology meaning.

EXAMPLE

A standards team has decided that their concept “Patient” should not mean “person receiving care.” They record that decision as a negative mapping because they want the knowledge base to preserve not only what the concept means, but also what it explicitly does not mean.

When they query the system, they see two kinds of results. First, they see concepts that are positively mapped to “person receiving care.” Those concepts represent a meaning that their own concept rejects, so they cannot be aligned with it for that ontology meaning. Second, they see other concepts that are also negatively mapped to “person receiving care.” That does not automatically mean those concepts are the same as theirs, but it tells the team that those concepts reject the same ontology meaning and may be worth comparing more carefully.

D3. As a standard owner, I want to start from a negative mapping and see both what my concept cannot align with and what other concepts reject that same ontology meaning.

This scenario concerns relation discovery starting from a standard concept that has a current negative mapping to an ontology concept. The negatively mapped ontology concept remains a valid point of reference for retrieval, but the interpretation differs from the positive case. Standard concepts that are positively mapped to that ontology concept represent a meaning that the seed concept explicitly does not take and therefore cannot align with it for that ontology meaning. Standard concepts that are also negatively mapped to the same ontology concept are relevant for further study because they reject the same ontology meaning, although they must not be treated as automatically aligned or semantically equivalent. Model concepts connected to the same ontology concept may also be retrieved and presented by polarity. This scenario is therefore concerned with direct same-ontology discovery from a negative starting point.

AS IS

A standard owner has a standard concept with a current negative mapping to an ontology concept and wants to see which standard concepts map positively to that ontology meaning and which other standard concepts also map negatively to it.

TO BE

The owner can retrieve positively mapped concepts that cannot align with the seed concept and negatively mapped concepts that may be worth further study, without treating them as automatically aligned.

D4. As a **model owner**, I want to know when my mapped concept has no related concepts in the knowledge base.

This scenario concerns the case in which a local model concept has a current mapping to an ontology concept but no other current concepts are connected to that ontology meaning, either directly or through inferred closure. In such a case, the absence of related concepts must not be interpreted as an error or as evidence that the mapping is invalid. Instead, the system should return an explicit empty result. This indicates that the concept is successfully mapped, but that no other model or standard concepts currently participate in the same semantic neighborhood. The scenario therefore formalizes the expected interpretation of a valid query that yields no related concepts.

AS IS

A model owner has a local concept with a current mapping to an ontology concept, but no other current mappings lead to related concepts for that ontology meaning, directly or indirectly.

TO BE

The owner learns that no related concepts were found for the current mapping.

D4. As a **model owner**, I want to know when my mapped concept has no related concepts in the knowledge base.

EXAMPLE

A model owner maps a local concept called “Rare patient category” to an ontology concept that seems correct for the intended meaning. After creating the mapping, they expect the system to show related concepts that might already exist elsewhere in the knowledge base.

Instead, the result is empty. No standard concepts and no other local concepts are currently connected to that ontology meaning, either directly or through the inferred closure that is available after reasoning. In this situation, the system should make it clear that the query worked correctly and that the empty result simply means there are no related concepts yet. This reassures the owner that the mapping is valid even though no reuse opportunities are currently visible.

D5. As a **standard owner**, I want to know when my mapped standard concept has no related concepts in the knowledge base.

This scenario concerns the case in which a standard concept has a current mapping to an ontology concept but no other current concepts are connected to that ontology meaning, either directly or through inferred closure. In such a case, the absence of related concepts must not be interpreted as an error or as evidence that the mapping is invalid. Instead, the system should return an explicit empty result. This indicates that the concept is successfully mapped, but that no other model or standard concepts currently participate in the same semantic neighborhood. The scenario therefore formalizes the expected interpretation of a valid query that yields no related concepts for a standard concept.

AS IS

A standard owner has a standard concept with a current mapping to an ontology concept, but no other current mappings lead to related concepts for that ontology meaning, directly or indirectly.

TO BE

The owner learns that no related concepts were found for the current mapping.

D5. As a **standard owner, I want to know when my mapped standard concept has no related concepts in the knowledge base.**

EXAMPLE

A standards team maps a new standard concept called “Rare care recipient” to an ontology concept that they believe is the right semantic anchor. After that, they query the system to see whether similar standard concepts or local concepts already exist.

The system returns no related concepts. That is not an error and it does not mean the mapping is wrong. It simply means that, at this point in the knowledge base, no other current concepts are connected to that ontology meaning. The system should make this explicit so the team understands that they may be the first to use that meaning in this particular modeling landscape.

I2. As a **user**, I want to start from an ontology concept and see everything currently connected to it.

This scenario concerns discovery that starts directly from an ontology concept rather than from a model concept or a standard concept. Because ontology concepts and mappings are the stored semantic entry points of the knowledge base, this is the most direct form of ontology-centered retrieval. The system should retrieve all current model concepts and standard concepts connected to the ontology concept and organize them by polarity. If reasoning has already expanded the graph, the result set should also include concepts whose relation to the ontology concept is inferred rather than directly asserted. The scenario therefore formalizes an ontology-centered view of the semantic neighborhood associated with a given ontology meaning.

AS IS

A user starts from an ontology concept and wants to see all current concepts connected to that ontology meaning.

TO BE

The user can retrieve all current model and standard concepts connected to the ontology concept, including concepts whose mapping to that ontology concept is inferred.

I2. As a **user**, I want to start from an ontology concept and see everything currently connected to it.

EXAMPLE

A user does not begin with a model concept or a standard concept. Instead, they start directly from an ontology concept, for example “person receiving care.” Their goal is to understand the full semantic neighborhood around that ontology meaning across the knowledge base.

The system retrieves all current local concepts and standard concepts that map positively to that ontology concept, as well as all concepts that map negatively to it. This already gives the user a direct ontology-centered view of what is compatible with that meaning and what is explicitly not compatible with it.

But the view does not have to stop there. Because the graph has already been expanded through reasoning, the system can also include concepts whose relation to that ontology concept is only visible indirectly. For example, a standard concept may become connected to “person receiving care” because another positively aligned standard concept propagates that mapping through the inference rules. The result is a richer ontology-centered picture that includes both directly asserted and indirectly inferred connections, while still keeping everything organized around the same ontology meaning.

I3. As a **standard owner**, I want to discover whether my concept can gain additional negative relations through ontology-based closure.

This scenario concerns the indirect derivation of a new negative relation for a standard concept through ontology-based closure. The seed concept already has mixed current mappings, including at least one positive mapping that connects it to another standard concept through shared ontology grounding. If two standard concepts share a positive mapping to the same ontology concept, alignment between them is inferred under the rule set. If one of those aligned concepts also has a negative mapping to another ontology concept, that negative mapping is propagated to the seed concept. Once that inferred negative mapping exists, the system can retrieve the concepts connected to the newly relevant ontology concept and organize them by polarity. The scenario therefore formalizes how additional negative semantic constraints can emerge indirectly from combinations of existing mappings and inference rules.

AS IS

A standard owner has a standard concept with mixed current mappings, including at least one positive mapping that connects it to another standard concept through shared ontology grounding, and wants to know whether an additional negative mapping can be inferred indirectly through the rule set.

TO BE

The owner can determine whether ontology-based closure produces a new negative mapping to another ontology concept and can inspect the related concepts that become reachable through that inferred negative mapping.

I3. As a **standard owner**, I want to discover whether my concept can gain additional negative relations through ontology-based closure.

EXAMPLE

A standards team is working with a concept called “Patient” that already has mixed current mappings. They have said that it should not mean “person receiving care,” but they have also said that it does mean “person in a service process.” They now want to know whether this combination can lead to additional negative knowledge through the inference rules.

The system finds another standard concept that also maps positively to “person in a service process.” Because the two concepts share that positive ontology grounding, the system can infer an alignment between them. It then sees that the second standard concept is negatively mapped to another ontology concept, such as “research participant.”

Because of that inferred alignment, the negative mapping to “research participant” is also propagated to the original concept. The result is that the original concept gains a new negative relation that was not asserted directly. The system can then open a new result branch for “research participant” and show the concepts connected to that ontology meaning, including positively mapped concepts that should not align with the seed concept for that meaning. At the same time, the original concept still has its earlier relations to the other ontology concepts. This is why the results must be presented by ontology concept rather than as one undifferentiated list.

L1. As a **model owner**, I want to understand what replaced an old mapping and use the current mapping or mappings instead.

This scenario concerns the interpretation of a historical local mapping that has been replaced by one or more newer current mappings. In this model, a mapping may have status replaced only when another mapping already exists and is explicitly linked to it through the queryable replaces relation. The replaced mapping must therefore be treated as historical rather than current. The system should expose the replacement path, identify all current replacement mappings, and use those replacement mappings as the basis for present-day discovery. Each replacement mapping may preserve or change the ontology concept, the polarity, or both. If more than one replacement exists, the resulting semantic neighborhoods should be presented separately for each replacement mapping.

AS IS

A model owner reaches an old mapping for a local concept. That old mapping may be positive or negative, and it has already been replaced by one or more newer current mappings.

TO BE

The owner can see that the old mapping is historical, identify the current replacement mapping or mappings, and use those current mappings for present-day discovery.

L1. As a **model owner**, I want to understand what replaced an old mapping and use the current mapping or mappings instead.

EXAMPLE

A model owner looks at an old mapping for the local concept “Patient” and sees that, in the past, it was linked to the ontology concept “person receiving care.” However, that mapping is no longer current. At some point, the team refined the meaning of the column and created newer mappings that better describe what the local concept now represents.

When the owner queries the old mapping, the system does not stop at telling them that the mapping is outdated. Instead, it follows the recorded replacement links and shows the newer mappings that replaced it. One replacement may point to a more appropriate ontology concept, while another may explicitly exclude an ontology concept that was previously considered. The owner can then use those newer mappings for current discovery, rather than relying on the historical one.

L2. As a **standard owner**, I want to understand what replaced an old mapping and use the current mapping or mappings instead.

This scenario concerns the interpretation of a historical standard mapping that has been replaced by one or more newer current mappings. In this model, a mapping may have status replaced only when another mapping already exists and is explicitly linked to it through the queryable replaces relation. The replaced mapping must therefore be treated as historical rather than current. The system should expose the replacement path, identify all current replacement mappings, and use those replacement mappings as the basis for present-day discovery. Each replacement mapping may preserve or change the ontology concept, the polarity, or both. If more than one replacement exists, the resulting semantic neighborhoods should be presented separately for each replacement mapping.

AS IS

A standard owner reaches an old mapping for a standard concept. That old mapping may be positive or negative, and it has already been replaced by one or more newer current mappings.

TO BE

The owner can see that the old mapping is historical, identify the current replacement mapping or mappings, and use those current mappings for present-day discovery.

L2. As a standard owner, I want to understand what replaced an old mapping and use the current mapping or mappings instead.

EXAMPLE

A standards team is examining an old mapping for the concept “Patient.” The old mapping is still visible in the history, but it is marked as replaced. This tells the team that the mapping is no longer the current one and that one or more newer mappings now describe the concept more accurately.

When they inspect the old mapping, the system follows the recorded replacement links and shows the newer mappings that superseded it. One replacement may point to a new ontology concept, while another may explicitly exclude a meaning that used to be associated with the concept. Instead of treating the old mapping as the basis for current discovery, the system uses the replacement mappings. This lets the team understand both the mapping history and the concept’s current semantic position.

L3. As a **standard owner, I want to detect when a conflict appears indirectly through the inference rules rather than from a directly asserted contradiction.**

This scenario concerns the detection of a contradiction that is not asserted directly but emerges through ontology-based inference. A standard concept may already have a current positive mapping to a given ontology concept and later acquire a negative mapping to that same ontology concept through the interaction of other current mappings and inference rules. This situation arises when the seed concept shares a positive ontology-grounded relation with another standard concept, allowing alignment to be inferred, and that aligned concept has a negative mapping to the ontology concept in question. The propagated negative mapping creates a polarity conflict on the same concept-ontology pair. In this situation, the pair must be treated as semantically unstable. The system should therefore expose the inferred conflict, make the derivation path explicit, and require governance review before that pair is used as a reliable basis for further discovery.

AS IS

A standard owner has a standard concept with current mappings that are individually valid, but together they can produce a contradiction through ontology-based inference.

TO BE

The owner can see that a conflict has been created indirectly, inspect the inference path that caused it, and treat that concept-ontology pair as unstable until it is reviewed.

L3. As a **standard owner, I want to detect when a conflict appears indirectly through the inference rules rather than from a directly asserted contradiction.**

EXAMPLE

A standards team has mapped their concept “Patient” positively to “person receiving care.” They have also mapped it positively to another ontology concept, such as “person in a service process.” At first, both mappings seem reasonable and there is no direct contradiction in what has been asserted.

Later, the system discovers another standard concept that also maps positively to “person in a service process.” Because of that shared positive grounding, the system infers that the two standards align. But that second concept also has a negative mapping to “person receiving care.” Through the inference rules, that negative mapping is propagated back to the original concept. Now the original standard has both a positive and a negative mapping to the same ontology concept. The system must therefore flag a conflict, explain how it was inferred, and prevent that concept-ontology pair from being treated as stable until someone reviews it.

C1. As a model owner, I want to retrieve related concepts for a set of mapped local concepts.

This scenario concerns relation discovery starting from a set of local model concepts rather than from a single seed concept. Each seed concept is expanded independently through its current mappings, and the retrieved results are grouped by ontology concept for that seed. After the per-seed expansion has been completed, the results may be combined into an aggregate view if needed. The same related concept may appear in multiple result groups if it is connected to more than one seed concept or if it is connected through more than one ontology concept. The scenario therefore formalizes how ontology-grounded discovery should be performed for multiple local concepts while preserving the semantic context of each seed.

AS IS

A model owner starts from a set of local concepts that already have current mappings and wants to retrieve related concepts for each seed concept and, if needed, across the set as a whole.

TO BE

The owner can retrieve related concepts for each starting concept in its own ontology-based context and can also combine the results across the whole set when a merged view is needed.

C1. As a model owner, I want to retrieve related concepts for a set of mapped local concepts.

EXAMPLE

A data steward is reviewing two local columns at the same time: one called “Patient” and another called “Study subject.” Each already has current mappings, but they point to different ontology concepts and therefore open different semantic neighborhoods. The steward wants to understand both concepts in one session without losing the distinction between them.

The system starts by expanding each seed concept separately. For “Patient,” it retrieves the concepts connected to the ontology concept “person receiving care.” For “Study subject,” it retrieves the concepts connected to “research participant.” Within each seed, the results are still grouped by ontology concept, so the steward can see exactly which meaning each group of related concepts comes from.

After that, the system can provide a combined view across both seed concepts. This merged view is useful for broader review, but it does not replace the per-seed view. A related concept may appear more than once if it is connected to both seed concepts or if it is connected through more than one ontology meaning. This helps the steward review multiple local concepts together without losing the semantic context of each one.

C2. As a **standard owner**, I want to retrieve related concepts for a set of mapped standard concepts.

This scenario concerns relation discovery starting from a set of standard concepts rather than from a single seed concept. Each seed standard concept is expanded independently through its current mappings and through any ontology-based closure supported by the rule set. The results are grouped first by seed concept and then by ontology concept, so that the semantic neighborhood of each seed remains explicit. After the per-seed expansions have been completed, the results may be combined into an aggregate view if required. The same related concept may appear in multiple result groups if it is connected to more than one seed concept or through more than one ontology concept. For positively mapped seed concepts, negatively mapped standard concepts to the same ontology concept are also relevant because they identify non-alignable cases under the rule set.

AS IS

A standard owner starts from a set of standard concepts that already have current mappings and wants to retrieve related concepts for each seed concept and, if needed, across the set as a whole.

TO BE

The owner can retrieve related concepts for each starting standard concept in its own ontology-based context, including non-alignable cases where relevant, and can also combine the results across the whole set when a merged view is needed.

C2. As a **standard owner**, I want to retrieve related concepts for a set of mapped standard concepts.

EXAMPLE

A standards team wants to review two standard concepts together, for example “Patient” and “Care receiver.” At first glance, the two look related, but the team wants a structured view of what surrounds each concept before deciding whether they overlap, complement each other, or should remain clearly distinct.

The system expands each standard concept separately. For one seed concept, it may show a branch around “person receiving care,” including positively related standards and local concepts. For the other, it may show a different ontology branch, such as “person receiving a service,” along with concepts that map negatively to that meaning and therefore should not align with the seed concept for that ontology concept.

After the per-seed expansion, the system can also provide a combined overview. That overview helps the team compare the two standards more efficiently, but the seed-by-seed and ontology-by-ontology structure remains important. The same related concept may appear in more than one result group if it is connected to both standards or through multiple ontology meanings. This makes the comparison more realistic and more useful for governance work.

C3. As a **model owner**, I want to compare two local concepts and see what they share and where they differ.

This scenario concerns the comparison of two local model concepts that already have current mappings. The purpose is not merely to merge their results, but to analyze semantic overlap and difference. The system should identify the ontology concepts shared by both seed concepts, the ontology concepts unique to each one, and the related concepts found under each ontology meaning. This makes it possible to determine whether the two local concepts are semantically close, partially overlapping, or clearly distinct. The scenario therefore formalizes ontology-based comparison between two local concepts.

AS IS

A model owner starts from two mapped local concepts and wants to compare what they share and where they differ.

TO BE

The owner can compare shared ontology meanings, ontology meanings unique to each seed concept, and the related concepts connected to each meaning.

C3. As a model owner, I want to compare two local concepts and see what they share and where they differ.

EXAMPLE

A data steward wants to compare two local concepts: “Patient” and “Study subject.” At first glance, the labels seem different, but the steward is not sure whether they overlap semantically or whether they should stay clearly separate.

The system compares the ontology meanings attached to both concepts. It shows which ontology concepts are shared by both, which ones are unique to only one of them, and which other local or standard concepts are connected through each ontology meaning. This helps the steward see whether the two local concepts are close in meaning, partially overlap in one area but diverge in another, or represent clearly distinct ideas that should remain separate in the model.

C4. As a **standard owner**, I want to compare two standard concepts and see what they share, where they differ, and where alignment is blocked.

This scenario concerns the comparison of two standard concepts that already have current mappings. As in the local comparison case, the goal is to identify shared ontology meanings, ontology meanings unique to each seed concept, and the related concepts found under each ontology meaning. In addition, standard-to-standard comparison must also highlight cases in which alignment is blocked because one concept maps positively and another maps negatively to the same ontology concept. The results should therefore remain grouped by ontology concept so that both overlap and incompatibility are made explicit. The scenario formalizes ontology-based comparison between two standard concepts, including the identification of non-alignable cases.

AS IS

A standard owner starts from two mapped standard concepts and wants to compare what they share and where they differ, including non-alignable cases.

TO BE

The owner can compare shared ontology meanings, ontology meanings unique to each seed concept, related concepts under each meaning, and standard concepts that cannot align because of opposite polarity.

C4. As a standard owner, I want to compare two standard concepts and see what they share, where they differ, and where alignment is blocked.

EXAMPLE

A standards team wants to compare two standard concepts, such as “Patient” and “Care receiver.” They suspect the concepts are related, but they are not sure whether they are close enough to align, only partially overlap, or should remain distinct because of important differences.

The system compares the ontology concepts attached to each standard concept and shows what is shared and what is unique. It also highlights standard concepts that cannot align because one maps positively and another maps negatively to the same ontology concept. This gives the team more than a simple similarity check. It shows exactly where the two standards overlap, where they diverge, and where alignment is blocked by opposite polarity, making the comparison much more useful for governance and harmonization decisions.